MAYANK

Basic C. Program (Unit 1)

1) Print Your name and Branch.

```
#include <stdio.h>
int main()
{
    printf("Name: Mayank Yadav");
    Printf("\n Branch: CSE(AI-M2);
    return 0;
}
```

2) Add two numbers entered By user

```
# include <stdio.h>
int main()
{
    int a, b;
    Scanf("%d %d", &a, &b);
    Printf("Sum = %d \n", a+b);
    return 0;
}
```

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3) Area of rectangle

```
# include < stdio.h>
int main ()
{
float length, breadth
scanf ("%f %f", &length, &breadth);
Printf ("Area = %.2f", length * breadth);
return 0;
}
```

Week - 2

1) Swap two No..

```
# include < stdio.h>
int main ()
{
int a, b, temp;
scanf ("%d %d", &a, &b);
temp = a;
a = b;
b = temp;
Printf ("a = %d \n b = %d\n", a, b);
return 0;
}
```

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2) Check even or odd

```
# include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    if (n%2 == 0)
        printf("Even");
    Else
        printf("odd");
    return 0;
}
```

input => 3

Output => Odd

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3) Simple Calculator

```
#include <stdio.h>
int main()
{
  double a, b;
  char op;
  scanf("%lf %c %lf", &a, &op, &b);
  if (op == '+')
    printf("%.2lf\n", a+b);
  else if (op == '-')
    printf("%.2lf\n", a-b);
  else if (op == '*')
    printf("%.2lf\n", a*b);
  else if (op == '/')
    {
      if (b != 0)
        printf("%.2lf\n", a/b)
      else
        print("division by zero not allowed");
    }
  else
    print("Enter valid operator");
  return 0;
}
```

1 If else

0 Even odd.

```
#include <stdio.h>
int main() {
    int =n;
    Scanf("%d", &n);
    in (n%.2 == 0)
        Print f("Even");
    else
        (Print f("Odd");
    return 0;
    }.
```

Input : 7
Output : odd

(+ve)(-ve) Zero

```
#include <stdion.h>
int main () {
    int n;
    scanf("%.d, &n):
    if (n>0)
        Print f("Positive");
    else if (n<0)
        Printf("Negative);
    Else print f("zero");
```

=) -5
=) Negati

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(3) Voting -

```
#include <stdio.h>
int main () {
  int age ;
    scanf ( "%d ", &age);
    if (age >= 18 )
        printf ( "Eligible.");
    else
        printf ("Not eligible ");
    return 0;
}
```

⇒ 20
⇒ eligible

4 Greatest of two Numbers.

```
#include <stdio.h>
int a, b ;
    scanf ("%d %d, &a, &b);
if (a>b)
    printf("%d", a);
else
    printf( "%d", b);
return 0;
}
```

input 10, 25
25

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(5) Greatest of Three No.

```
#include <stdio.h>
int main () {
    int a, b, c.
    Scanf ("%d%d%d", &a, &b, &c);
if (a>b && a>c)
    printf ("%d", a);
else if (b>c)
    printf ("%d", a);
else
    printf ("%d", b);
else printf ("%d", c);
    return 0;
}
```

10, 11, 12

=) 12

(6) Pass or fail

```
include <stdio.h>
int main () {
Scanf ("%d", &m);
if (m>=33)
    printf ("pass");
Else printf ("Fail");
return 0;
}
```

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Leap Year

```
#include <stdio.h>
int main() {
    int y;
    scanf("%d", &y);
    if (y%400==0 || (y%4==0 && y%100!=0))
        Print("Leap Year")
    else
        printf("Not Leap Year");
    return 0;
}
```

Vowel and Consonant:

```
#include
int main() {
char ch;
Printf("Enter Three Characte:")
Scanf("%c", &ch);
if (ch='a' || ch='e' || ch='i'
    ch='o' || ch='u')
printf("vowel\n");
else printf("consonant\n");
return 0;
}
```

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9/ Grade System

```
include <stdio.h>
int main() {
   int m;
   Scanf("%d:", &m);
  if (m>=90) Printf("A")
  else if (m>=75) Printf("B");
      "    (m>=60) "("C");
      "    (m>=33) "(D);
   else print if ("fail");
   Return 0;
}
```

Electricity

```
# include <stdio.h>
int main () {
float bill, units
Scanf ("%f%f", &units, &bill);
if (units >200)
    bill = bill * 0.80;
else if (units >100)
    bill = bill * 0.90.
print ("%2f", bill)
    return 0;
}
```

250 1000

Output = 800.00

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Switch Case

1) Simple Calculator

```
# include < stdio.h>
  int main ( ) {
int ~~float~~ a, b; char o;
    Scanf(" %d %c %d ,&a,&o ,&b
    switch(o) {
    case '+' : printf ( " %d" , a+b) ; break ;
    case '-' printf ( " %d" ,a-b) ; break;
    case '*' printf ("%d" , a*b.) ; break
    case '/' printf .( "%d., a/b);
      }}
```

2) Day

```
# include < stdio.h>
int main ( ) {
  int n;
    Scanf("%d", &n);
  switch (n) {
  case 1 : printf ("mon") : Break;
  Cas 2: pust f ("Tue") : Brak;
  Case 7  pust f ('Sun") : ;
  }}
```

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Area

```
# include < stdio.h.
  int main () {.
int  n, a, b : scanf ("%d , &n);
     Switch .(n) {..
(Circle) case I : Scanf ("%d", &a) : printf ("%..2f"
                                  , 3.14*.a*a);
(Square) break : .case 2 .scanf ("%.d , &a); printf ("%d.
                                  , a*a) : break;
(Rectangle) case 3 : scanf ("%.d%.d., &a, &b.) ; printf ("%d' a
     }}.
```

Traffic

```
# include <stdio.h>.
  int main () {
  char c; scanf ("%.c", &c);
    Switch (c) {
       Case 'R' :printf ( " Stop") ; Break;.
       Case 'Y' :printf ("Ready") ; Break ;
       Case : 'G' : printf ("Go") ;
    }}
```

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Student details

```
#include <stdio.h>

Struct Student
{
    int roll no;
    char name[50];
    float marks;
};
int main()
{
    Struct student s;
    Printf("Enter Roll no _");
    Scanf("%d", &s.rollno);
    Printf("Enter name :");
    Scanf("%49[^\n]", s.name);
    Printf("Enter mark:")
    Scanf("%f", &s.marks);
    Printf("\n Student details\n");
    Printf("Roll Number : %d\n", s.rollno);
    Printf("Name : %f\n", s.marks);
    Return 0;
}
```

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~~Operators~~

if else calculator

```
#include <stdio.h>
int main()
{
    float a, b;
    char op;
    printf("Enter expression -");
    scanf("%f %c %f", &a, &op, &b);
    if (op == '+') printf("Answer = %.2f \n", a+b);
    else if (op == '-') printf("Answer = %.2f \n", a-b)
    else if (op == '*') printf("Answer = %.2f \n", a*b)
    else if (op == '/' b != 0) printf("Answer = %.2f \n", a/b
    else print("Invalid input \n");
    return 0;
}
```

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Types of loop

(i) For loop

(ii) do while loop

(iii) while loop.

(a) For loop.

For Initialization, Condⁿ, Updation.

```
for (int i=1; i<=20; i++) {
    printf("%d\n", i);
```

i++ Post Increment

i-- post decrement

(b) while loop

while (condⁿ) is true.

```
}
while (i<11)
{
    printf("The value of i is %d\n", i);
    i++;
}
```

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WAP to sum of even and odd

```
#include <stdio.h>
int main() {
    int i, sumeven = 0, sumodd = 0
    for (i=1; i<=500; i++) {
    {
        sumeven += i
    } else {
        sumodd += i
    }
    printf("sum of even num: %d\n", sumeven);
    printf("sum of odd nu: %d\n"; sumodd
```

Que: WAP in go to topic

```
#include <stdio.h>
int main() {
    int i = 1
    Start:
    printf("%d\n", i);
    i++;
    if (i<=10)
    goto Start;
    return 0;
}
```

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Que - WAP a no is odd or even using ternary operator..

```
#include <stdio.h>
int main("enter the No:");
Scanf("%d", &n);
Printf(n%2==0 ? "even" : "odd.");
```

in 3 check which is greator

```
#include<stdio.h>
int main()
{ int a, b, c, largest;
Printf("enter the No");
Scanf("%d %d %d", &a, &b, &c
largest = (a>b) ? ((a>c) ? a : c) : ((b>c
? (b : c);
Printf("The large No is: %d, larg);
return 0;
}
```

Output: Enter 3 No, 1 2 3

The larg No = 3

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Leap Year

```
if ((n%4 == 0) && (n%100 != 0)||(n%400 == 0))
{
    printf("%d is the leap Year", n);
}
```

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Q-1

Answer

```
#include <stdio.h>
int main() {
  // statement
  return 0;
}
```

Answer - 2

Compiler	Interpreter
Translate all once whole Program	Translate line by line
Faster Execution	slower execution
Example C, C++	Example, Python, JS

Ans - 3

while ⇒ Condition checked before execution
may execute 0 times

do while ⇒ Condition checked after execution
executes at least once.

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④ If else ladder used to check multiply conditions.

```
if (condition)
    statement ;
else if (condition2)
    statement ;
else
    statement ;
```

⑤ output

```
KGEC ! Hello
```

Section B

Answer 6

1) Preprocessing : Handles #include, #define.
2) Compilation : Converts C code to assembly
3) Assembly : Converts assembly to object code
4) Linking : Links libraries and object files
5) Execution : Program runs and produces output.

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Answer 7

Primary data type in C.

		Example
Char	1 byte	'A', 'B', —
int	4 bytes	10, 11, —
float	4 bytes	2.5, 3.6678
double	8 bytes	2.567589
Void	No storage	No value

Range = ~~(4bytes-1)~~ $\Rightarrow 2^{(8\text{bytes}-1)}$

Answer 8

Implicit automatically By Compiler

int a = 5; float b = a;

Explicit = Done by programmer.

float b = (float) 5/2;

Program
```
#include <stdio.h>
int main() {
  int a, b, c;
  scanf("%d%d%d", &a, &b, &c);
  printf("Average = %.2f", (a+b+c)/3.0);
  return 0;
}
```

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Answer. 9

Sum of even no

```
include <stdio.h>
int main() {
   int n, i, sum = 0;
  scanf("%d", &n);
  for (i=2 ; i<=n ; i+=2)
       sum += i;
  Printf("sum = %d\n", sum);
  Printf("%s", sum>50 ? "Greater than 500"
       : "Not greater than 50");
  return 0;
}
```

Prime numbers from 1 to n

```
#include <stdio.h>
int main() {
 int n, i, j, prime;
 scanf("%d", &n);
 for(i=2; i<=n; i++) {
  prime = 1;
  for (j=2; j*j <= i; j++)
       if (i%j == 0) {prime = 0; break;}
 if(prime) Printf("%d", i); }
       return 0;
}
```

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Section C

1.) Operators in C

-1) Relational: Compare Values and return 0/1

<, >, <=, >=, ==, !=

Example a > b

2)

Logical: Combine Conditions

&& AND, || OR, ! NOT

Example: a > 0 && b > 0

3) Bitwise

work on Individual

&, |, ^, ~, <<, >>

Example: 5 & 3 = 1.

Largest of two no.

```
#include <stdio.h>
int main() {
  int a b max
  scanf("%d %d", &a, &b);
  max = (a > b) ? a : b;
  printf("Largest = %d", max);
return 0;
```

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12) Loops in C

Loop	Condition	Min^n execution
for	Before	0 times
while	Before	0 times
do while	After	1 time

Syntax

```
for(int ; condition ; update { }
while(condition) { }
do { } while(condition);
```

Pallindrome Program

```
#include <stdio.h>
int main() {
  int n, t, r=0;
  scanf("%d", &n); t=n;
while(n){ r = r*10 + n%10; n/=10; }
printf( t==r ? "Palindrome" : "Not
Pallindrome");
  return 0;
}
```

Section A

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Answer 1

Constant ⇒ Can't changed by the programmer while the code is in execution. or been done.

Variable ⇒ Can be changed by the programmer while the code is in execution.

Answer 2

Structure

```
# include <stdio.h>
int main()
{
    // Your code
return 0;
}.
```

Answer 3

if	if else
For single condition	For more than or any backup condition
if (condition) {code}	if (true case) { else (false case) . }

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Answer 4

Purpose of Switch statement

use to choose or execute a single one Block of the code from multiple alternative path based on any single value or variable in code.

Answer 5

The any code on any example in which code runs Infinitely in any particular sequence in the given condition.

Example printing counting No from 0 to ∞.

Section B

Basic Structure of C.

```
# include <stdio.h>
int main ( )
{
    // Your Code
return 0;
}
```

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Steps

Writing a code

↳ Preprocessing

↳ Compilation

↳ Addressing

↳ Linking

↳ Execution

Example

```
#include <stdio.h>
int main()
{
printf("Hello World")
return 0;
}
```

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Answer - 7

Type conversion

There are two type of conversion

implicit type	Explicit type
It is automatically done By the Compiler	It is manully done by the Programmer
Ex int a a = 5 float a a = 10.56	float b = float (b/2);

Type conversion Required to convert the value in C programing. like int to float or vice versa.

Answer 8

The Input and Output statement are

Input = scanf ("%d" &a);

Output = printf (" print %d , a);

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Example

```
# include <stdio.h>
int main ()
{
int a, b;
scanf ("%d %d", &a &b);
printf ("Sum = %d", a+b);
return 0;
}
```

Answer 9

Grade

```
# include <stdio.h>
int main ()
{
int marks;
scanf ("%d", &marks);
if (m >= 90)
    printf ("A");
if else (m >= 75)
        printf ("B");
if else (m >= 60)
    Printf ("C");
else
    printf ("D");
return 0;
}
```

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Pgm 10 Sum upto 100

```
#include <stdio.h>
int main()
{
  int i, sum=0;
for (i=1; i<=100; i++)
   sum = sum+i;
printf("sum=%d", sum);
 return 0;
}
```

Pgm 11

Operator in C

- Arithmetic : + - * / %
- Relational : < > <= >= == !=
- Logical : && || !
- Assignment : = , += , -= , *=
- Increment / decrement : ++ , --
- Bitwise : & | ^ ~ << >>
- Ternary : ? :

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Example

```
#include <stdio.h>
int main() {
  int a = 10, b = 5;
  printf("%d %d %d %d", a+b, a>b, a&&b);
  return 0;
}
```

Qn 12

Break Terminate the loop

Continue Skips current iteration

goto Jumps to labelled/Given statement

Ex Infix order

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```
#include <stdio.h>

int main()
{
int i;
for(i=1; i<=5; i++)
{
  if(i==2)
    continue;   //Skip order
  if(i==5)
    break;   // Stop after 4 orders.
    printf("order %d processed\n", i);
}
goto done;
done:
  printf("completed");
  return 0;
}
```

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1D Array in C -

Declaration, Input, Output, Transversal

Theory (100 words)

collection of elements of same data typed stored in contigous memory locations array.

Indexing starts from 0

Declaration for Initialization

```
int arr[5]
int a[5] = {12,45,7,23,10};
```

Input output and transversal

Declaration = defines an array and its size

Input : Reads value using Scanf()

Transversal : Visits each element using a loop

Output : Display value using printf()

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Array operation

Max, Min, Sum, Avg

Example = {12, 45, 7, 23, 10}

Maximum = 45

Minimum ~~12~~ 7

Sum = 12 + 45 + 7 + 23 + 10 = 105

Avg = 105 / 5 = 21

```
int arr[5] = {12, 45, 7, 23, 10};
int max = arr[0], min = arr[0], sum = 0;
for(int i = 0; i < 5; i++)
{
    if(arr[i] > max) max = arr[i];
    if(arr[i] < min) max = arr[i];
}
float avg = (float) sum / 5;
Printf("Max = %d - Min = %d\n", Max, min);
Printf("Sum = %d. Avg = %.2f", sum, avg);
```

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Important -

Initialize max and min ordars[0]

So the program also works correctly when array elements are negative.

String in C:

A string is sequence of characters terminated by the null character ('\0').

string are stored in character array

```
Char * str[100];
```

```
Scanf("%s", str);
```

```
fgets(str, sizeof(str), stdin);
```

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Functions	Purpose
Strlen(str)	Return String Length
Strcpy(dst, src)	String copy Karna.
Strcmp(s1, s2)	do String compare Karna
Puts(str)	String print Karna

(1)

```
Char str[] = "Hello"
Printf("%u", strlen(str)); // 5
```

(2)

```
Char s1[20] = "Hello";
Chas s2[20];
Strcpy(s2, s1);
Printf("%.s", s2); //Hello
```

(3)

```
Printf("%d", strcmp("apple", "apple"));
```

// 0

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Strcmp() Result;

- 0 : String = equal Hain

Negative : pehli string lexicographically smaller hai

Positive : pehli string greater hai

Note Strlen() '\0' Ko count Nhi Karta

3 pointers in C

Pointer EK variable hai jo Kisi doosre variable Ka memory address store karta hai

& : variable Ka address deta hai.

* : pointer Ko dereference karke us address par store value deta hai

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Output →

a = 10

p ⟶

Ex

```
int main() {
int a = 10;
int *p = &a
Printf("value = %d \n", a);
Printf("address = %p \n", (void*)p);
Printf("Value using pointer = %d \n", *p);
return 0;
}
```

Output

Value = 10

address = (memory address)

Value using pointer = 10

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da

WAP Update two value using pointers

~~# Stdio.h~~

```
# include <stdio.h>
void update (int *a, int *b)
{
    int sum = *a + *b;
    int diff = *diff - *b;
    if (diff < 0)
        diff = -diff;
    *a = sum;
    *b = diff;
}
int main () {
    int a, b;
    scanf ("%d %d" &a, &b);
    update (&a, &b);
    Printf ("%d \n %d \n", a, b);
    return 0;
}
```

Input
=10, 4
Output
16, 6

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Important points

⇒ A pointer stores an address not a actual value of the pointed to variable

⇒ &a gives the address of a.

⇒ *p access the value of the address stored in p.

④ File Handling in C

File handling is used to create read, write and manage data stored in files it allows data to remain available after a program ends

File handling are provided by the header file <stdio.h>

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Important functions

Function	Purpose
fopen()	Opens a file
fclose()	Closes a file
fprintf()	Writes formatted data to file
fscanf()	Reads formatted data from a file
fgets()	Reads a line from file
fputs()	Writes a string to file

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Program

```
#include <stdio.h>
int main() {
    File *fp;
    char name[50];
```

Declar Variables

```
    fp = fopen("data.txt", "w");
```

for writing → open

```
    if (fp = NULL) {
        printf("File Not opened \n")
        return 1;
    }
```

check opened or not

```
    printf("Enter name and age:");
    scanf("%49s %d", name, age);
```

take input

```
    fprint(fp, "%s %d \n", name, age);
```

data writing

```
    fclose(fp) // Close file
```

Real life Example:

Used In student record system payroll management storing user data log files configuration files etc..

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// Reading

```
FP = fopen ("data.txt", "r")

if (fp==NULL)
        {printf ("No file \n");
         return 1;
        }

fscanf (fp, "%49s %d", name, &age);
printf ("Data from file: \n");
printf ("Name: %s, age: %d \n", name, age);
fclose (fp);

return 0;
}
```

Read and Display saved data

Close the file

Output Enter name and age: MynR 21

Data from File:

Name: MynR, Age: 21

Python

(*) Unit - 04

Python: It is a high level, interpreted, general-purpose Programming language known for its simple & readable syntax.

- created by Guido Van Rossum in 1989, 1991.

- **Variables**: It is a name used to store a value in a program.

ex → name = "Rahul"
Age = 18
marks = 85.5

output → Rahul
18
85.5

(*) name, age, marks → Variable

- **Data types**:
  It specifies the kind of value stored in a variable.

(i) int → int Num
(ii) float → Decimal Num
(iii) Str → Text
(iv) bool → True / False

(v) list → Ordered, changeable collection
(vi) tuple → " , unchangeable collection
(vii) Set → collection of unique values
(viii) Dict → key-value pairs.

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→ Type Conversion

(i) int ( ) → convert a suitable value in integer.

(ii) float ( ) → convert a suitable value in decimal num.

(iii) str ( ) → convert a value to a string.

(iv) bool ( ) → convert a value into a boolean.

→ Operators

(a) Arithmetic operator → +|-|*|/|//|%|**

ex →
10+5 = 15
10-5 = 5
10*5 = 50
10/5 = 2.0

10//5.
11//2 → 5
11%2 → 1
2**3 → 8.

(b) Comparison operator
↳ ==|!=|>|<|>=|<=

ex ⇒ a=10, b=20

print (a==b)	output
print (a!=b)	False
print (a<b)	True
	True

(c) logical operator :
↳ and | or | not

(d) Assignment operator :
↳ =|+=|-=|*=|/=

(*) Python control flow

(a) If/else :-

(i) If statement : executes a block of code when condn is true.

(ii) If-else statement : select between two alternatives

(iii) If-else-elif statement : multiple condn in sequence.

ex:- Syntax

```
If condn :
    statement
```

```
If condn :
    // code
else :
    // code
```

```
If condn :
    // statement
elif condn :
    // statement
else :
    // statement
```

Code :

```
a = 10
b = 25
c = 15
if a>=b and a>=c:
    print ("A is greatest")
elif b>=a and b>=c
    print ("B is greatest")
else :
    print ("C is greatest")
```

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(b) **for loop** : It repeats a block code for each item in a sequence.

Syntax : for (variable) in (sequence) :
statement

code : for i in range(1,6) :
print(i)

output :
1
2
3
4
5

code-2 for i in range(1,11);
print(5,'x',i,"=",5*i)

→ output
5x1=5
5x2=10
;
5x10=50

(c) **while loop** :
It repeats a block of code as long as its condn remains true.

Syntax : while (condn) :
statement.

code : i=1
while i<=5;
print(i)
i+=1

output :
1
2
3
4
5
6

CamScanner

Code

(i) If-else / If-else-elif

(a) Even / odd check :

```
n = int(input("enter a num"))
if n % 2 == 0:
    print("Even")
else:
    print("ODD")
```

Output:
n=5
odd
n=4
Even.

(b) Positive / negative / Zero

```
n = int(input("Enter a num:"))
if n > 0:
    print("Positive")
elif n < 0:
    print("negative")
else:
    print("Zero")
```

Output
n=6
Positive
n=-4
negative.

(c) Pass / Fail

```
marks = int(input("enter mark"))
if marks >= 33:
    Print("Pass")
else:
    Print("Fail")
```

marks = 54
Pass

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(d) Greater of two Num:

```
a = int(input("enter 1st num"))
b = int(input("enter 2nd num"))
if a>b:
    print(a, "is Greater")
elif b>a:
    print(b, "is Greater")
else:
    print("both are equal")
```

(*) a=6
b=4
6 is greater

(e) Eligible to Vote:

```
age = int(input("enter age"))
if age>=18:
    print("Eligible to vote")
else:
    print("Not eligible")
```

(*) age = 19
Eligible to vote

(f) leap year or not

```
y = int(input("Enter year:"))
if y%400==0 or (y%4==0 and y%100!=0):
    print("leap year")
else:
    print("Not a leap year")
```

(*) y = 2028
leap year

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(g) Small / capital letter

```
ch = input("enter a letter:")
if ch.isupper():
    print("capital letter")
elif ch.islower():
    print("small letter")
else:
    print("Not an alphabet")
```

@ch
enter a letter : C
capital letter

(h) Discount system

```
amt = float(input("enter shopping amount :"))
if amt >= 5000:
    d = 20
elif amt >=3000:
    d=10
elif amt >=1000:
    d=5
else:
    d=0
print("Discount =", amt*d/100)
print("final amount =", amt*(100-d)/100)
```

enter a amount : 1728
discount = 86.4
final amount = 1641.6

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① Grade System (A, B, C, D)

```
marks = int(input("enter marks:"))
if marks >= 80:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
elif marks >= 40:
    print("Grade C")
else:
    print("Grade D")
```

```
Enter marks : 64
Grade B.
```

② Username & password check.

```
u = input("enter username:")
p = input("enter password:")
if u == "Deepanshu" and p == "281706":
    print("login successful")
else:
    print("wrong username & password")
```

```
① Enter username : Deepanshu
enter Password : 2806
    Invalid username or
        password.
```

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→ For loop

① Num from 1 to N.

~~for i in num~~

```
N = int(input("Enter num"))
for i in range(1, N+1):
    print(i)
```

enter Num: 10

1
2
3
4
5
6
7
8
9
10

② Print table

```
n = int(input("enter a num:"))
for i in range(1,11):
    print(n, "x", i, "=", n*i)
```

enter a num: 2

2 x 1 = 2
2 x 2 = 4
⋮
2 x 10 = 20

③ Sum of first 'N' number

```
n = int(input("enter a num:"))
s = 0
for i in range(1, n+1):
    s = s + i
print("sum=", s)
```

enter a num = 10
Sum = 55

④ Print even num:

```
for i in range(2,51,2):
    print(i)
```

2
4
6
8
10
⋮
50.

CamScanner

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⑤ Factorial of a num :

```
n = int(input("enter a no:"))
f = 1
for i in range(1, n+1):
    f *= i
print("factorial =", f)
```

Enter a no : 6
factorial = 720

⑥ Count vowels in string

```
s = input("enter string")
c = 0
for i in s:
    if ch.lower() in "aeiou":
        c += 1
print("Vowels =", c)
```

Enter a string : Deepanshu
Vowels = 4

⑦ Reverse a string

```
s = input("enter string:")
rev = ""
for i in s:
    rev += i
print("reverse =", rev)
```

Enter a string : SINGH
reverse = HGNIS

⑧ Print list element

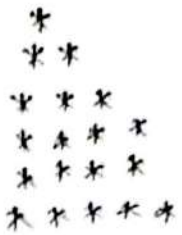
```
a = [17, 28, 100, 101]
for i in a:
    print(i)
```

17
28
100
101

CamScanner

⑨ Print star pattern

```
for i in range (1,6):
    print ("*"*i)
```



⑩ Find sum of all elements in list

```
a = [17,28,30,40,50]
s = 0
for i in a:
    s += i
print ("sum =", s)
```

sum = 165

(*) Discount on shopping more than 5000:

```
amt = float (input ("enter purchase"))
if amt >=5000
    discount = amt * 0.10
    final_amount = amount - discount
    Print ("discount 10%)
else:
    discount = 0
    final_amount = amount
    Print ("No discount")
Print ("original amount :", final_amount)
```

Enter purchase = 6789
discount 10%
final amount = 6110.1

CamScanner

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ATM machine

```
balance = 1000.0
choice = 0
while choice != 4:
    print("\n---- ATM MENU ----")
    print("1. check balance")
    print("2. Deposit money")
    print("3. withdraw money")
    print("4. Exit")
    choice = int(input("Enter your choice: "))
    if choice == 1:
        print("current balance:", balance)

    elif choice == 2:
        amt = float(input("Enter deposit money"))
        balance += amt
        print("deposit successful.")

    elif choice == 3:
        amt = float(input("enter withdrawal money:"))
        if amt <= balance:
            balance -= amt
            print("withdraw successful")
        else:
            print("Insufficient balance")

    elif choice == 4:
        print("Thank you for using ATM. Good bye!")

    else:
        Print("Invalid choice")
```

CamScanner

Program 1

Print Name and Branch

```
#include <stdio.h>
int main() {
    printf("Name : Mayank Yadav\n");
    printf("Branch: CSE AI & ML\n");
    return 0;
}
```

CamScanner

Input

—

Output

Name : Mayank Yadav

Branch : CSE (AI & ML)

CamScanner

MAYANK

Program 2

Add two No entered by user

```
# include <stdio.h>
int main ( ) {
    int a, b;
    Scanf ("%d %d", &a &b);
    printf (" %d %d"  a+b );

    return 0;
}
```

CamScanner

Input

10 20

Output

30

CamScanner

Program 3

Area of Rectangle

```
#include <stdio.h>
int main () {
    float length, breadth;
    scanf("%f %f", &length , &bread
    printf("Area = %.2f" = length * bre
    return 0;
}
```

CamScanner

MAYANK

Input

5 3.5

Output

17.50

CamScanner

MAYANK

Program 4

Swap two no

```
#include <stdio.h>
int main() {
    int a, b, temp;
    scanf("%d %d", &a, &b);
    temp = a;
    a = b;
    b = temb;
    printf("a = %d \n b = %d", a, b);
    return 0;
}
```

CamScanner

Input

5 . 10

Output

a = 10

b = 5

CamScanner

MAYANK

Program 5

Check even or odd

```
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    printf(n%2==0 ? "Even : odd");
    return 0;
}
```

CamScanner

Input :

7

Output :

Odd

CamScanner

MAYANK

Program 6

Simple Calculator

```
#include <stdio.h>
    int main() {
        int a, b;
        char op;
        scanf("%d %c %d", &a, &op, &b);
        if(op == '+') printf("%d", a+b);
    else if(op == '-') printf("%d", a-b);
    else if(op == '*') printf("%d", a*b);
    else if(op == '/') printf("%d", a/b);
        else printf("Invalid operator");
        return 0;
    }
```

CamScanner

MAYANK

Input

10 + 5

Output

15.00

CamScanner

MAYANK

Program 7

Largest of 3 Number

```
# includ <stdio.h>
int main() {
    int ,a, b ,c, max ;
    Scanf ("%.d %d %.d ", &a &b &c);
    max = a ;
    if (b > max)  max = b ;
    if (c > max) max = c ;
    prinf ("Largest = %.d", max );
    return 0 ;
}
```

CamScanner

MAYANK

Input

12 45, 30

Output

Largest = 45

CamScanner

Program 8

Grade Calculator

```
#include <stdio.h>
int main() {
    int marks;
    scanf("%d", &marks);
    if (marks >= 90) printf("Grade A");
    else if (marks >= 80) printf("Grade B");
    else if (marks >= 70) printf("Grade C");
    else if (marks >= 60) printf("Grade D");
    else //("print F
        printf("Grade F");
    return 0;
}
```

CamScanner

Input

84

Output

Grade B

CamScanner

Program 9

Day Name

```
# includ <;stdio.h>
int main() {
  switch (day) {
      Cas 1 : prin ("monday") ; break;
      Cas 2 ; prin ("Tusday") ; break;
      Cas 3; prin ("Wednesday) ; break;
      Cas 4; prin (" Thursday) ; break;
      Cas 5 : prin (" Friday") ; break;
      Cas 6; prin (" Saturday") ; break;
      Cas 7; prin ("Sunday") ; break;
      default : print ("Invalid day") ;
  }
  return 0;
}
```

CamScanner

Input

3

Output

Wednesday

CamScanner

MAYANK

Program - 10

Multiplication table of N.

```
#include <stdio.h>
int main () {
    int n;
    scanf("%d", &n);
    for (int i = 1; i <=10; i++)
        printf("%d x %d = %d \n",
                n, i, n*i);
    return 0;
}
```

CamScanner

Input

5

Output

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50

CamScanner

Program 11

Sum of First N Natural Numbers

```
#include<stdio.h>
int main () {
    int n , sum = 0;
    scanf ("%d", &n );
    for (int i =1 ; i <=n ; i++).
        sum +=i ;
    printf ("sum =%d" , sum);
    return 0;
}
```

CamScanner

Input

5

Output

Sum = 15

CamScanner

MAYANK

Program 12

Count digits in a Number

```
#include <stdio.h>
int main() {
    int n, count = 0;
    scanf("%d", &n);
    if (n < 0) n = -n;
    do {
        count++
        n /= 10;
    } while (n > 0)
    printf("Digits = %d", count);
    return 0;
}
```

CamScanner

MAYANK

Input

54326

Output

digit: 2

CamScanner

MAYANK

Program - 13

Prime numbers between 1 to 50

```
#include <stdio.h>
int main() {
    for (int n = 2 ; n <= 50 ; n++) {
        int prime = 1;
        for (int i = 2 ; i * i <= n; i++)
            if (n % i == 0) {prime = 0; break;
            }
        if (prime) printf ("%d", n);
    }
    return 0;
}
```

CamScanner

MAYANK

Input

—

Output

2, 3 5 7 11 13 17 19 23

29 31 37 41 43 47

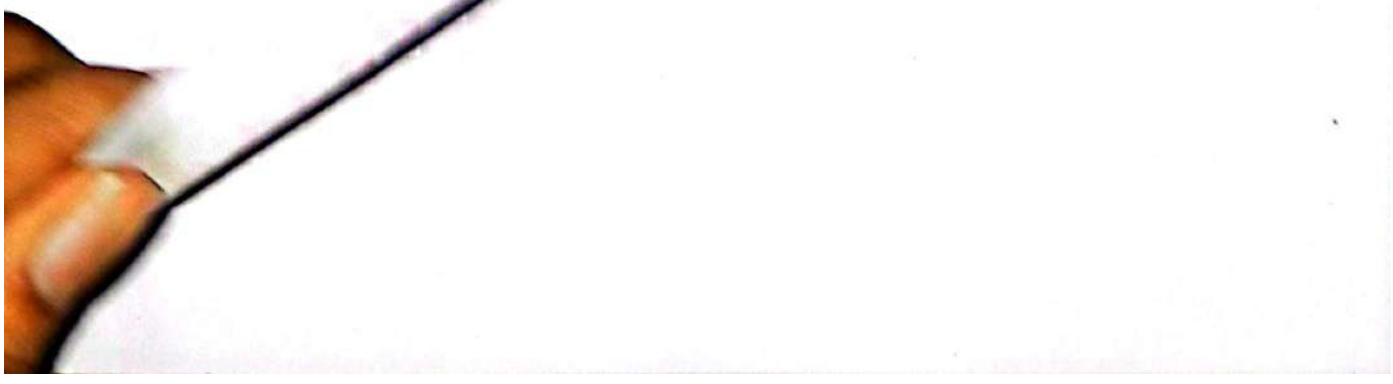
CamScanner

MAYANK

Program 14

Fibonacci Series.

```
#include <stdio.h>
int main() {
    int a = 0, b = 1, c;
    for (int i = 0; i < 10; i++)
    {
        printf("%d ", a);
        c = a + b;
        a = b;
        b = c
    }
    ret
}
```



CamScanner

MAYANK

Input

—.

Output

0 1 1 2 3 5 8 13, 21 . 34

CamScanner

MAYANK

Program - 15

Reverse a Number

```
# include <stdio.h>
int main ( ) {
        int n, rev = 0;
      Scanf ("%.d;" &n);
      while (n != 0) {
        rev = rev * 10 + n % 10;
        n /= 10;
    }
    printf ("Reverse = %.d", rev);
    return 0;
}
```

CamScanner

MAYANK

Input

1 2 3 4

Output

4 3 2 1

CamScanner

MAYANK

Ans- Program - 16

Factorial using function

```
#include <stdio.h>
    int fact(int n) {
        int f = 1;
        for (int i = 1; i <= n; i++)
            f *= 1;
        return f;
    }
    int main() {
        int n;
        scanf("%d", &n);
        printf("%d", fact(n));
        return 0;
    }
```

CamScanner

MAYANK

Input

78687 5

Output

Palindrom Factorial : 120

CamScanner

MAYANK

Program 17

Check palindrome

```
#include <stdio.h>
int isPalindrome(int n) {
    int rev = 0, temp = n;
    while (temp > 0) {
        rev = rev * 10 + temp % 10;
        temp /= 10;
    }
    return rev == n;
}
int main() {
    int n;
    scanf("%d", &n);
    printf(isPalindrome(n) ? "Palindrome"
        "Not Palindrome");
    return 0;
}
```

CamScanner

MAYANK

Input

121

Output

Palindrome

CamScanner

MAYANK

Program - 18

Find GCD of two numbers

```
#include <stdio.h>
int main() {
    int a, b, temp;
    Scanf("%d%d", &a, &b);
    while (b != 0) {
        temp = b;
        b = a%b;
        a = temp;
    }
    printf("GCD = %d", a);
    return 0;
}
```

CamScanner

Input

72 8

GCD

8

CamScanner

MAYANK

Program 19

Recursive factorial

```
#include <stdio.h>
long long factorial (int n) {
    if (n <= 1) return 1;
    return n * factorial (n-1)
}
int main() {
    int n;
    scanf ("%d", &n);
    printf ("%lld", factorial(n));
    return 0;
}
```

CamScanner

MAYANK

Input

5

Output

120

CamScanner

Program 20

Recursive Fibonacci

```
#include<stdio.h>

int fabioncci(int n) {
    if(n <= 1) return n;
    return fibonacci(n-1) + fibonacci(n-2);
}
int main() {
    int n;
    Scanf("%d", &n);
    Printf("%d", fibonacci(n));
    return 0;
}
```

CamScanner

Program 20

Recursive Fibonacci

```
#include <stdio.h>

int fabioncci (int n) {
    if (n <= 1) return n;
    return fibonacci (n-1) + fibonacci (n-2);
}
int main () {
    int n;
    Scanf ("%.d", &n);
    Printf ("%.d", fibonacci (n));
    return 0;
}
```

CamScanner

Input
⇒ 7

Output
⇒ 13

CamScanner

MAYANK

Program 21

Sum of digits

```
#include<stdio.h>
int SumDigit(int n) {
    if(n==0) return 0;
    return n%10 + SumDigits(n/10);
}
int main() {
    int n;
    Scanf("%d", &n);
    printf("Sum= %d", SumDigits(n));
    return 0;
}
```

CamScanner

MAYANK

Input

1234

Output

Sum = 10

CamScanner

Program 22

Find Max Min

```
#include<stdio.h>
int main() {
    int n, a[100];
    scanf("%d", &n);
    for (int i=0; i<n; i++)
    scanf("%d", &a[i]);
    int max = a[0], min = a[0];
    for (int i=1; i<n; i++) {
        if (a[i] > max) max = a[i];
        if (a[i] < min) min = a[i];
    }
    printf("maximum = %d \n minimum = %d",
                max, min);
    return 0;
}
```

CamScanner

MAYANK

Input

5

10 3 35 7 1

Output

Maximum = 25

Minimum = 1

CamScanner

MAYANK

Program 23

Average

```
#include <stdio.h>
    int n, a[100];
        float sum = 0;
        scanf ("%d", &n);
        for (int i = 0; i < n; i++) {
            scanf += a[i];
            Sum += a[i];
        }
        printf ("Avg = %.2f", sum/n);
        return 0;
    }
```

CamScanner

MAYANK

Input

5

10 20 30 40 50

Output

Average = 30.00

CamScanner

MAYANK

Program 24

Linear Search in Array

```
#include <stdio.h>

int main() {
 int n;
 Printf("Enter number of element:\n");
  Scanf("%d", &n)
  int arr[n]
  Printf("Enter %d elements : \n", n)
  For(int i = 0; i < n ; i++)
      Scanf("%d", &a[i]);
      Scanf("%d", &Key);
      for(int i=0; i<n; i++)
      if(a[i] == Key) { found i ; break ; }
  if(found != -1) Printf("Found at index %d"
  else Printf("Not found") found);
  return 0;
}
```

CamScanner

MAYANK

Input

5

10 20 30 40 50

30

Output

Found at index 2.

CamScanner

Program 25

Count Vowels

```
#include <stdio.h>
#include <strin.h>
int main() {
    Char str[100];
    int count = 0;
    Scanf("%.[^\n]", str);
    for(int i = 0; str[i]; i++)
        if(strch("aeiouAEIOU", str[i]))
            count++;
    Printf("Vowels = %.d", count);
    return 0;
}
```

CamScanner

MAYANK

Input

Hello World

Output

Vowels = 3

CamScanner

Program 26 Reverse a String

```
#include <stdio.h>
#include <string.h>
int main () {
    char str[100], temp;
    scanf("%[^\n]", str);
    int len = strlen(str);
    for(int i = 0; i < len/2; i++){
        temp = str[i];
        str[i] = str[i];
        str[i] = str[len-1-i];
        str[len-1-i] = temp;
    }
    printf("%s", str);
    return 0;
}
```

CamScanner

MAYANK

Input

hello

Output

olleh

CamScanner

MAYANK

Program 27

Check string is palindrome

```
#include <stdio.h>
#include <string.h>
int main() {
    char str[100];
    int len, palindrome = 1;
    scanf("%s", str);
    len = strlen(str);
    for (int i = 0; i < len/2; i++).
        if (str[i] != str[len - 1 - i])
    { palindrome = 0; break; }
printf(palindrom ? "palindrom" : "Not");
    return 0;
}
```

CamScanner

Input

madam

Output

Palindrom

CamScanner

Program 28

Store 5 Numbers

```
# include <Stdio.h>

int main ( ) {
        int a[5];
    for (int i = 0; i < 5; i++)
        Scanf ("%d", i < 5; i++)
    For (int i = 4; i >= 0; i--)
        printf ("%d", a[i]);
    return 0;
}
```

CamScanner

Input

1 2 3 4 5

Output

5 4 3 2 1



CamScanner

MAYANK

Program 29

String length, copying, concatenation

```
#include <stdio.h>
#include <string.h>
int main() {
    char s1[100], s2[100], copy[100];
    scanf("%s %s", s1, s2);
    printf("length = %d\n", strlen(s1));
    strcpy(copy, s1);
    printf("copied = %s\n", copy);
    strcat(s1, s2);
    printf("concantenated = %s", s1);
    return 0;
}
```

CamScanner

MAYANK

Input

Hello World

Output

Length = 5

Copied = Hello

Concatenated = HelloWorld

CamScanner